

Virtual reality and noise canceling headphone distraction during pediatric dermatologic procedures

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Abstract

Pain and anxiety related to medical procedures have long been recognized as a significant healthcare concern. If a patient's procedural pain and anxiety are not addressed, long-term physical and psychological sequelae including increased perceived pain, anxiety, disruptive behavior, trauma reactions, or refusal of future procedures can occur. The objective of our study was to assess the utility of a virtual reality (VR) headset or noise-canceling headphones (HP) compared to treatment as usual (TAU) in reducing pain and anxiety during pediatric dermatology procedures. Results indicated a significant difference between pre- versus post-procedure anxiety in the VR and HP groups but not the TAU group suggesting non-pharmacologic technology-based interventions such as VR and headphones may reduce patients' anxiety during pediatric dermatology procedures.

KEYWORDS

dermatology, distraction, pediatrics, procedure, psychology, virtual reality

1 | INTRODUCTION

Pain and anxiety related to medical procedures may lead to increased pain, anxiety, disruptive behavior, trauma reactions, and refusal of future procedures.¹ Anesthesia and other pharmacologic options used to manage pediatric pain and anxiety during procedures include risks associated with anesthesia, longer procedure times, greater use of resources, and increased costs. Thus, non-pharmacologic options can be beneficial. Virtual reality (VR) (Figure 1) and other technology-based interventions have been shown to reduce patient anxiety and pain in several studies within pediatric dentistry, cancer, wound care, and laceration repair.^{2–5} However, there is limited research

examining technology-based behavioral interventions within pediatric dermatology.⁶

2 | METHODS

Following institutional review board approval, children undergoing a procedure were recruited from our pediatric dermatology clinic. Procedures included excimer laser, laser for hair removal or vascular lesions (long pulsed Nd:YAG; pulsed dye laser), intramuscular or intralesional injections, excision, biopsy, incision and drainage, and cryotherapy. Inclusion criteria was English- or Spanish-speaking patients aged 5–18 years old.

Exclusion criteria were children with a history of a seizure disorder, audiovisual challenges preventing adequate use of electronics, lasers involving the face where VR would make it difficult for appropriate eye protection. Patients were randomized to one of three groups: treatment as usual (TAU), noise-canceling headphones (HP), or VR. Depending on the procedure, all patients regardless of randomization group were offered standard of care including topical anesthetics, cold spray, or vibration device as well as non-pharmacological measures such as comfort holds and caregiver support. Pre- and post-procedure pain and anxiety scores were reported by both the patient and caregiver using the Numerical Rating Scale (NRS-11)⁷ for pain and the Visual Analog Scale-Anxiety (VAS-A)⁸⁻¹⁰ for anxiety. The patient reported their subjective feelings regarding pain and anxiety, while the caregiver reported what they thought the patient was experiencing.

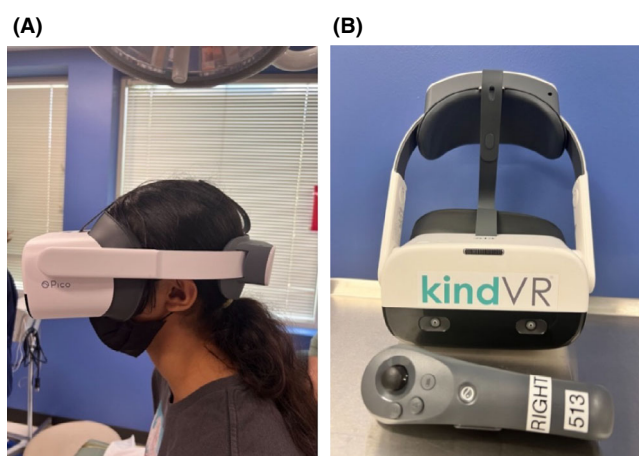


FIGURE 1 Virtual reality headset.

3 | RESULTS

A total of 109 patients were included of which 35% were male and 65% were female, and racial and ethnic groups included 48% Hispanic, 26% White, 14% Black, 9% Other, 3% Asian. Forty-three patients (40%) were randomized to TAU, 32 (29%) to HP, and 34 (31%) to VR.

Pairwise comparison between patients' pre- and post-procedure anxiety scores showed a statistically significant decrease in patients' anxiety in the HP group compared to TAU (mean difference 2.289; 95% CI (0.318, 4.258); $p = .017$). Paired sample t -tests, which assessed whether the pre- and post-procedure anxiety scores were significantly different within each treatment group, showed a significant difference in both the VR and HP groups but not the TAU group. Thus, participants who used VR or HP had significantly decreased patient anxiety scores post-procedure, whereas anxiety scores for participants who received TAU did not change significantly. Statistically significant differences were present in patient-reported pain pre- and post-procedure in all treatment groups (see Table 1), indicating that the procedures were painful. It is worth noting that despite all groups showing statistical significance, changes in pain scores did vary across the groups. This finding was mostly corroborated by the caregiver-reported pain (see Table 2). Caregiver-reported anxiety was significantly lower in all groups.

4 | DISCUSSION

These findings suggest non-pharmacologic technology-based interventions such as VR and headphones can be helpful in reducing patient anxiety in pediatric dermatology procedures. Most

TABLE 1 Patient-reported anxiety and pain scores (paired sample t -test).

Condition	Mean anxiety difference (pre-post)	T	p	95% CI	Mean pain difference (pre-post)	T	p	95% CI
Virtual reality	1.757	2.793	.009*	0.476, 3.039	-2.2441	-3.627	.000**	-3.181, -1.072
Treatment as usual	-0.186	-0.349	.729	-1.261, 0.889	-2.581	-5.827	.000**	-3.475, -1.687
Headphones	2.581	5.516	.000**	1.625, 3.536	-1.281	-2.368	.024*	-2.384, -0.178

* $p < .05$. ** $p < .001$.

TABLE 2 Caregiver-reported anxiety and pain scores (paired sample t -tests).

Condition	Mean anxiety difference (pre-post)	T	p	95% CI	Mean pain difference (pre-post)	T	p	95% CI
Virtual reality	1.774	2.99	.006*	0.562, 2.986	-1.907	-3.01	.005*	-3.196, -0.616
Treatment as usual	1.049	2.145	.038*	0.061, 2.037	-2.537	-4.953	.000**	-3.572, -1.501
Headphones	2.571	3.301	.003*	.973, 4.170	-0.742	-1.104	.278	-2.114, 0.630

* $p < .05$. ** $p < .001$.

importantly, participants' decrease in anxiety in the two groups using behavioral distractions (VR and HP), but not the TAU group, indicates that behavioral distractions may be beneficial in lowering patient anxiety, which can linger even after a procedure's completion. Findings from our study do not overwhelmingly support one technology-based intervention over the other. Instead, they suggest that the use of either intervention can provide benefits. From a logistical and cost perspective, although HP are less novel and exciting, they may be easier for clinics to obtain and offer to youth during procedures. The significant increase in pain across all groups indicates that the procedures were indeed painful, which unfortunately due to the small sample size, we could not control for. Caregivers' perceptions of pain and anxiety were not completely aligned with their child's perceptions. Caregivers felt all treatment groups were beneficial in decreasing anxiety. This difference may be best explained as reporting on others' experience of pain and anxiety is often subjective and may reflect the anxiety of the caregiver, which was not assessed in this study. Of note, although this was a one-time study for each participant, providers who performed repeat procedures reported that patients who previously participated in the study were more willing to undergo a repeat procedure (i.e., repeat cryotherapy for verruca vulgaris) if they could have access to the technology-based interventions used in this study during clinic.

Limitations of this study include having multiple procedure types and dermatologic diagnoses that vary with respect to time and expected pain or anxiety; and some participants received repeat procedures while others were receiving them for the first time. Additionally, future studies should assess if there are positive or negative practice effects around frequent use of VR, such as increasing compliance with future procedures.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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